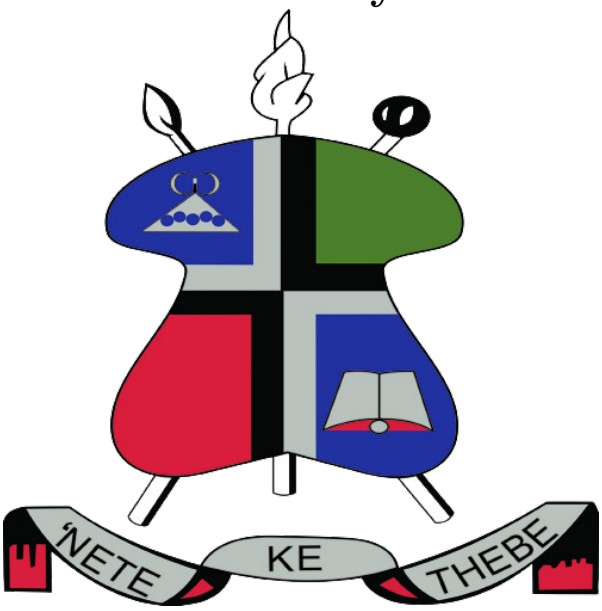
National University of Lesotho



Project Proposal Draft:

Detection and Classification of Lung Cancer using Deep Learning

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1 Theoretical Background

Lung cancer is one of the leading causes of cancer-related deaths globally[8]. Computed tomography (CT) imaging is the primary tool used by radiologists to identify pulmonary nodules that may indicate malignancy, enabling clinicians to visualize internal thoracic structures in cross-sectional slice images[7]. However, manually evaluating high-resolution CT scans is time-consuming and subject to inter-observer variability, where different radiologists reach different conclusions on the same scan[7].

Deep Learning (DL) architectures, specifically Convolutional Neural Networks, have proven to be ideal for visual feature extraction in image processing[7][4]. Convolutional Neural Networks (CNNs) are considered effective for automatically learning hierarchical spatial patterns such as edges, textures and basic shapes, then complex structures like lung nodule shapes directly from pixel data.

2 Problem Statement

In limited resource healthcare centers, such as referral facilities in Lesotho like Queen 'Mamohato Memorial Hospital, there is lack of specialized radiologists and operational Computed Tomography (CT) scanners[5]. As a result, diagnostic waiting times for CT scan interpretations are long and early-stage pulmonary nodules can be overlooked due to manual visual review under heavy daily workloads. Delay diagnosis likely leads to late-stage cancer detection and lowers patient survival rates[8].

3 Motivation

Automating the detection and classification of lung cancer using Convolutional Neural Networks provides a fast, objective decision-support mechanism for healthcare professionals evaluating thoracic CT scans[2]. Furthermore, this project serves as an Industrial Solution (IS) that aligns with national priorities, including Lesotho's National Digital Transformation Strategy and the Lesotho Digital Health Strategy, which improves public service delivery to enhance healthcare quality and accessibility.

The project will help meet healthcare professionals halfway by helping in diagnosis so that the number of casualties are reduced due to misdiagnosis in early lung cancer detection.

4 Aim and Objectives of the Project

4.1 Aim

To develop and evaluate deep learning models for classifying lung CT scans as Normal, Benign or Malignant using the IQ-OTH/NCCD dataset, and to deploy the best-performing model in a web-based application for use by medical professionals.

4.2 Objectives

1. To prepare the IQ-OTH/NCCD lung CT dataset for model training through preprocessing, augmentation and splitting into training, validation and test sets.
2. To develop and train Convolutional Neural Network (CNN) and a pre-trained ResNet50 fine-tuned through transfer learning.
3. To evaluate and compare the two models using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices, and select the best-performing model.
4. To deploy the selected model in a web-based application that allows medical professionals to upload a CT scan and receive a classification result.

5 Scope of the Project

5.1 In-Scope:

- Slice-level (2D) CT image preprocessing and classification (Normal/Benign/Malignant) using a Convolutional Neural Network[7].
- Grad-CAM based visual explainability overlays for each prediction[4, 6].
- De-identification of identifying metadata fields from uploaded scans prior to storage.
- A web-based prototype with upload, prediction and heatmap display.
- Model evaluation (accuracy, precision, recall, F1, ROC-AUC and confusion matrix).

5.2 Out-of-scope:

- 3D volumetric CT analysis using CNN.
- Clinical deployment and integration with hospital PACS systems.

6 Methodology

6.1 Overview

The project follows an experiment-driven pipeline: dataset acquisition and exploration, preprocessing, data splitting and augmentation, model training on the preprocessed dataset, classification into Normal, Benign and Malignant, Grad-CAM explanation, evaluation, and integration into a web application.

6.2 Dataset Acquisition and Description

Table 1: Summary of the IQ-OTH/NCCD lung cancer CT dataset.

Attribute	**Description**
Dataset name	IQ-OTH/NCCD
Source institution	Iraq-Oncology Teaching Hospital / National Center for Cancer Diseases
Year collected	2019
Availability	Publicly available on Kaggle
Data type	CT scan slices (JPEG images)
Total images	1097
Total cases	110
Privacy	All images de-identified before analysis

Table 2: Class distribution of cases in the IQ-OTH/NCCD dataset.

Class	**Cases**	**Images**	**Share of Cases(%)**
Normal	55	416	50.0
Malignant	40	561	36.4
Benign	15	120	13.6
Total	**110**	**1097**	**100.0**

6.3 Preprocessing

1. **Format handling:** Only PNG/JPEG files are accepted. And uploaded DICOM files are read and converted to PNG images.
2. **De-identification:** Patient-identifying DICOM tags (name, ID, birth date and similar fields) are removed before conversion or any storage.
3. **Intensity conversion(DICOM only):** Raw values are rescaled to standard CT units and a lung window is applied before conversion to PNG, so that lung tissue and nodules keep adequate contrast.

4. **Resizing:** All images are resized to the input size required by the model.

5. **Grayscale conversion:** Colour carries little diagnostic information in CT, so images are converted to grayscale. Then the grayscale slice is replicated across three channels.

6. **Normalization:** Pixel values are scaled to the fixed range.

7. **Patient-level splitting**: The data is split into training, validation and test sets, stratified by class. All slices from one patient must be placed in the same subset. Since the dataset does not provide patient identifiers. Consecutive images whose similarity exceeds a chosen threshold are placed in the same group, and the split is done by group. Otherwise, near-identical adjacent slices appear in both training and test data and inflate the reported performance. The test set is evaluated only once, after model selection has been finalised.

8. **Data Augmentation (training set only)**: Small random rotations, translations and zoom are applied to mimic natural differences in patient positioning. Horizontal flips and mild blur are tested cautiously and kept only if they improve validation performance.

9. **Class imbalance**: The dataset is imbalanced at slice level (Malignant 561, Normal 416, Benign 120; largest-to-smallest ratio about 4.7:1). This is handled on the training set only, using class-weighted loss or weighted sampling.

6.4 Feature Extraction and Mapping

In the transfer-learning model, convolutional layers of ResNet50 shall be used to examine lung CT slices and extract complex patterns since ResNet50 has already learned visual features from ImageNet[4]. The ResNet50 classification layer shall be removed since it was designed for classifying ImageNet categories rather than Normal, Benign and Malignant classification. The global average pooling shall convert the extracted information into a feature vector of 2048 values for each slice since the final convolutional block has 2048 maps of 7×7. Then the feature vector shall be passed to our newly added classification head, consisting of a dense layer with ReLU activation, dropout and a 3-way softmax layer to classify each CT slice into Normal, Benign or Malignant categories. In the first training stage, the convolutional base of ResNet50 is frozen so that only the newly added classification layers are trained.

The extracted feature vectors will also be projected to two dimensions using PCA and t-SNE, for visualization only[1]. These plots show whether Normal, Benign and Malignant slices form separable clusters and where misclassified slices lie. This helps diagnose confusion between the Malignant, Benign and Normal classes.

6.5 Model Development & Training

Two models shall be developed and compared: a baseline CNN trained from scratch and a ResNet50 transfer-learning model. The baseline quantifies the benefit of ImageNet pre-training on a small, imbalanced medical dataset[4]. Both models take a 224×224 input (the grayscale slice replicated across three channels) and output a probability for each of the three classes (Normal, Benign, Malignant).

6.5.1 Baseline CNN

The baseline shall consist of four convolutional blocks. Each block applies a 3×3 convolution with ReLU activation, batch normalisation and 2×2 max pooling. The number of filters doubles at each block (32, 64, 128 and 256) and the spatial size is halved from 224×224 down to 14×14 while the number of feature maps grows with the number of filters. The final feature maps of block 4, with 256 maps of 14×14, shall be passed through global average pooling to form a feature vector of 256 values where each value represents a feature map. The dropout is set to 0.5 to deactivate 50% of neurons to force the network to rely on multiple paths and prevent overfitting but at prediction time, the dropout shall be switched off. The output layer shall have 3 neurons, each for Normal, Benign and Malignant, which receives the 256 values. Each neuron shall compute a weighted sum of the 256 values plus a bias and the softmax function converts the three sums into probabilities that add up to 1. The class with the highest probability (confidence score) is the predicted class. Pixel values shall be normalized to be between $[0, 1]$ by dividing by 255. The network shall be trained from scratch with the Adam optimiser at an initial learning rate of about 10^{-3}.

6.5.2 ResNet50 Transfer Learning Model

ResNet50 uses residual (skip) connections that address the vanishing gradient problem and make very deep networks easier to train[3, 4]. The ResNet50 model uses ImageNet weights with the original classification layer removed, which gives a $7 \times 7 \times 2048$ feature map. This is passed to a new classification head consisting of global average pooling (a feature vector of 2048 values), a dense layer with ReLU activation, dropout (0.5) and a 3-way softmax layer.

Training has two stages:

1. **Feature extraction:** the convolutional base shall be frozen and only the head shall be trained, with a learning rate of about 10^{-3}.

2. **Fine-tuning:** the final convolutional block shall be unfrozen and trained at a much lower learning rate of about 10^{-5}, while the early layers stay frozen. Batch normalisation layers remain frozen.

6.5.3 Training Configuration

Both models shall use class-weighted categorical cross-entropy to handle class imbalance[7]. Adaptive learning rates for head training and fine-tuning shall be 10^{-3} and 10^{-5} respectively. Batch size shall be between 16 and 32 so that batch fits in memory. Number of Epochs shall be between 30 and 50 so the model calculates the prediction error using the loss function and the weights using backpropagation and Adam optimizer[1]. The training shall stop when validation loss stops improving for a set number of epochs and restore the best weights to prevent overfitting and save time. When validation loss stops decreasing, the learning rate shall be multiplied by the factor of between 0.1 and 0.5.

6.6 Grad-CAM Explainability

Grad-CAM shall be computed at the last convolutional layer of ResNet50 (conv5_block3_out)[4, 6]. Using TensorFlow's GradientTape, the gradient of the predicted class score is computed with respect to that layer's feature maps. These gradients are global-averaged-pooled to obtain a weight for each channel. The feature maps are combined as a weighted sum, passed through a ReLU, upsampled to the input size, and overlaid on the original scan as a heatmap.

Grad-CAM heatmaps are coarse, show where the model looked but not why, and can create false confidence[6]. To check them, the following steps are taken:

- Heatmaps for correct and incorrect predictions are compared, and inspected for whether the model attends to lung tissue rather than image borders, the scanner table, or annotations.
- A sample of heatmaps is reviewed by the supervisor or a clinician, if available.

6.7 Model Evaluation

The following metrics shall be computed on the held-out test set:

- accuracy, per-class precision, recall (sensitivity), F1-score and specificity, with macro and weighted averages;
- one-vs-rest ROC-AUC per class, and the confusion matrix.

Because a missed cancer is the costliest error, Malignant recall is the priority metric[2]. Since the test set is small, 95% confidence intervals are computed by bootstrap resampling (1,000 resamples)[1]. Further checks are:

- **Comparisons:** baseline CNN versus ResNet50, with and without denoising and class weighting.
- **Robustness:** k-fold cross-validation, grouped by patient, on the training data.

- **Error analysis:** misclassified slices are inspected, especially Benign/Normal confusion, using the Grad-CAM overlays.
- **System evaluation:** inference time per slice, functional testing against FR1–FR10, and user acceptance testing with the supervisor and, if possible, a clinician.

6.8 Web Application Development and Integration

An Agile, prototype-driven approach is used across three layers, in line with the system design:

- **Backend (FastAPI):** loads a versioned model, applies the same preprocessing as in training, runs inference, generates the Grad-CAM overlay, and returns the class, confidence score and heatmap. Every prediction request is logged.
- **Database (PostgreSQL):** stores user credentials, de-identified metadata, prediction logs (input reference, output, confidence, model version, timestamp), and clinician feedback (accept, reject or modify).
- **Frontend (HTML, CSS, Javascript):** provides login, upload, a side-by-side view of the original scan, heatmap overlay and classification result, and an administrator view of model performance.

Security includes authentication, role-based access, and encryption in transit and at rest. Testing covers unit tests for preprocessing and the API, end-to-end integration tests, and user acceptance testing.

7 Detailed System Requirements

7.1 Functional Requirements

FR1: The System shall allow a registered medical professional to log in and access the diagnostic dashboard.
FR2: The System shall allow upload of a CT scan (either in DICOM or PNG/JPEG formats).
FR3: The System shall preprocess and normalize the uploaded scan (resizing, windowing, artifact removal) before reaching the conclusion.
FR4: The System shall run the trained CNN model on the preprocessed scan and return a classification (Normal, Benign, Malignant) with a confidence score.
FR5: The System shall generate a Grad-CAM heatmap overlay highlighting the regions that drove the classification.
FR6: The System shall display the original scan, the heatmap overlay, and the classification result side-by-side in the web UI.
FR7: The System shall log every prediction (input reference, output, confidence, model version, timestamp) for auditing and later evaluation.

FR8: The System shall train a Convolutional Neural Network and evaluate it, reporting accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix[7].
FR9: The System shall anonymize or remove data that identifies the patient from the metadata before processing and storage.
FR10: The System shall allow an administrator to view model performance over time.

7.2 Non-functional Requirements

Performance: The System shall build an inference which will be within few seconds per CT slice.
Usability: The System shall have a clean User Interface (UI) that enables the medical professional to navigate the dashboard with good User Experience (UX) without machine learning expertise.
Security: The system shall have authentication, role-based access, and encryption when storing or transmitting patient data.
Constraint: The System shall allow handling of corrupted or unsupported upload formats.
Feedback and Explainability: The System shall provide an explanation for every prediction and allow the medical professionals to provide feedback (accept, reject or modify the prediction) to support continuous model evaluation.

8 High-level/Conceptual System Design

8.1 Presentation Layer

- Clinician web application where the end users shall interact with the System.
- Upload form (single image or DICOM series).
- Results view (detection, classification, confidence level and heatmap overlay side by side).
- The administrator shall monitor the model performance metrics of the System.

8.2 Application Layer

- Preprocessing module: Handles resizing, normalization and windowing.
- CNN inference engine: loads the trained model from storage and returns classification with confidence scores[4].

- Grad-CAM module: produces spatial heatmap showing which regions of the upload influenced the prediction[4, 6].
- Logging and metrics service: records what happens in the pipelines, in the form that can be queried.

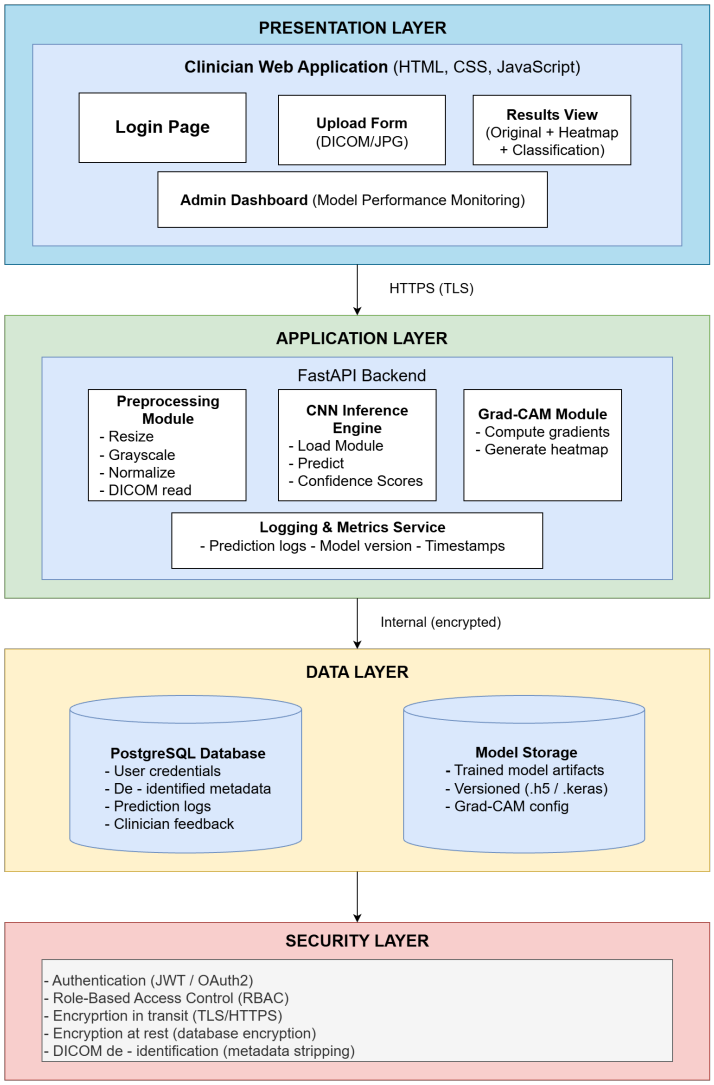
8.3 Data Layer

- Database: stores imaging metadata, de-identified clinical data, model prediction logs and clinical credentials. A relational schema (PostgreSQL) shall be used given the structure and nature of the data.
- Model Storage: stores trained model artifacts, each identified by a version.

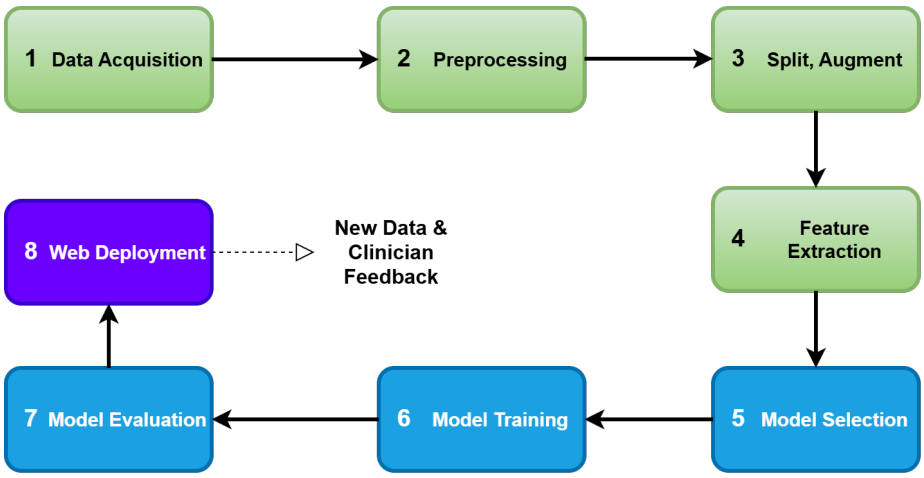
8.4 Security Layer

The security layer shall be responsible for enforcing OSI security services across the presentation, application and data layer.

8.5 System Architecture



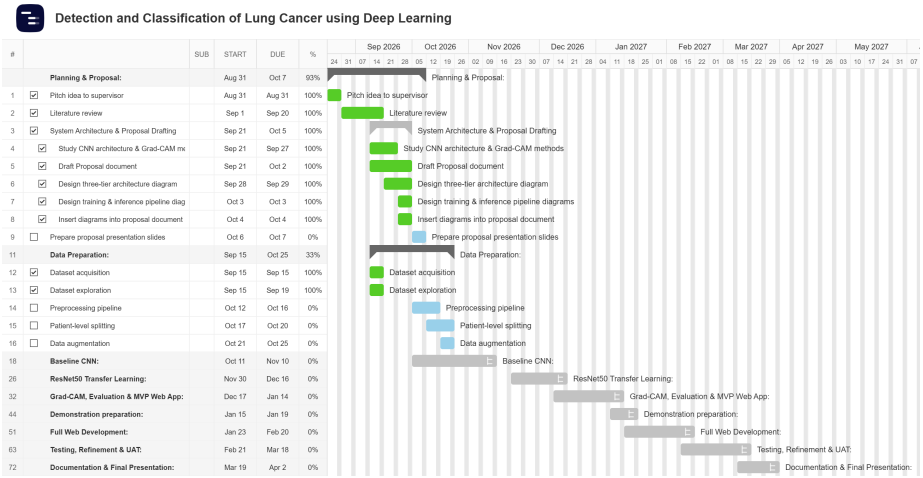
8.6 Machine Learning Pipeline



9 Tools and Technologies to use

Category	**Tool/Technology**
Language	Python
Deep Learning Framework	TensorFlow
Data handling and Image Processing	NumPy, Pandas, OpenCV, Pydicom
Model architectures	Custom CNN; transfer learning option: ResNet50
Backend	FastAPI
Frontend	HTML, CSS and JavaScript
Database	PostgreSQL
Dataset	IQ-OTH/NCCD
Version control	Git/GitHub
Environment	Jupyter Notebook

10 Project Schedule (Estimate Roadmap)



#		SUB	START	DUE	%
	Planning & Proposal:		Aug 31	Oct 7	93%
1	☑ Pitch idea to supervisor		Aug 31	Aug 31	100%
2	☑ Literature review		Sep 1	Sep 20	100%
3	☑ System Architecture & Proposal Drafting		Sep 21	Oct 5	100%
4	☑ Study CNN architecture & Grad-CAM m		Sep 21	Sep 27	100%
5	☑ Draft Proposal document		Sep 21	Oct 2	100%
6	☑ Design three-tier architecture diagram		Sep 28	Sep 29	100%
7	☑ Design training & inference pipeline diag		Oct 3	Oct 3	100%
8	☑ Insert diagrams into proposal document		Oct 4	Oct 4	100%
9	☐ Prepare proposal presentation slides		Oct 6	Oct 7	0%
11	**Data Preparation:**		Sep 15	Oct 25	33%
12	☑ Dataset acquisition		Sep 15	Sep 15	100%
13	☑ Dataset exploration		Sep 15	Sep 19	100%
14	☐ Preprocessing pipeline		Oct 12	Oct 16	0%
15	☐ Patient-level splitting		Oct 17	Oct 20	0%
16	☐ Data augmentation		Oct 21	Oct 25	0%
18	**Baseline CNN:**		Oct 11	Nov 10	0%
26	**ResNet50 Transfer Learning:**		Nov 30	Dec 16	0%
32	**Grad-CAM, Evaluation & MVP Web App:**		Dec 17	Jan 14	0%
44	**Demonstration preparation:**		Jan 15	Jan 19	0%
51	**Full Web Development:**		Jan 23	Feb 20	0%
63	**Testing, Refinement & UAT:**		Feb 21	Mar 18	0%
72	**Documentation & Final Presentation:**		Mar 19	Apr 2	0%

11 Conclusion

This proposal has outlined a plan to design, implement, and evaluate a Convolutional Neural Network (CNN) integrated into a web application for detecting and classifying lung CT scans into Normal, Benign and Malignant categories. The project addresses the shortage of specialised radiologists and CT scanners in Lesotho[5] by providing a fast, objective decision-support tool with Grad-CAM explainability.

A two-model approach is adopted: a baseline CNN trained from scratch and a ResNet50 transfer-learning model, compared on the IQ-OTH/NCCD dataset. Class-weighted cross-entropy handles the 4.7:1 class imbalance, while patient-level splitting and adjacent slice detection prevent data leakage. Grad-CAM highlights the regions that drive each prediction, and evaluation uses accuracy, precision, recall, F1-score, ROC-AUC and the confusion matrix, with Malignant recall as the priority metric.

The system is delivered as a three-tier web application using FastAPI, PostgreSQL, and HTML/CSS/JavaScript. A minimum viable product (MVP) will be demonstrated in January 2027, and the full system, including database, authentication, DICOM handling and clinician feedback, will be completed and tested by April 2027. Clinical deployment and PACS integration are out of scope; the project delivers a working prototype to demonstrate feasibility.

References

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[3] Kaiming He et al. "Deep Residual Learning for Image Recognition". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90.

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[8] World Health Organization. *Lung Cancer*. 2026. URL: https://www.who.int/news-room/fact-sheets/detail/lung-cancer (visited on 09/02/2026).